Spatial Degradation of Hepatocyte Zonation in Human MASLD

An analysis report: building, validating and transferring a zonation ruler

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1 Background and question

The liver lobule is spatially organized along a porto-central axis: pericentral hepatocytes (near the central vein) and periportal hepatocytes (near the portal triad) run complementary metabolic programs, and most hepatocyte genes are *zonated*. **Paper 2** (Yakubovsky & Itzkovitz, *Nature* 2026) built a spatially resolved atlas of the *healthy* human liver and a quantitative zonation reference. **Paper 1** (Gribben, Vallier et al., *Nature* 2024, GSE202379) profiled ∼69k hepatocytes by single-nucleus RNA-seq across 47 donors spanning MASLD stages (Healthy → NAFLD → NASH → Cirrhosis → End-stage), but is *spatially blind*. Paper 1 reports qualitatively that "hepatocytes lose their zonation".

This project makes that quantitative. We use Paper 2's healthy zonation as a *ruler* to assign each Paper 1 hepatocyte a pseudo-zonation coordinate, then test, at the level of the **donor** (never the cell, to avoid pseudoreplication):

- **H1**: does zonation collapse with disease stage?
- **H2**: which gene programs lose their zonal restriction (slope), and fastest?
- **H3**: are de-zonated cells also more plastic (hepatocyte↔cholangiocyte)?

2 Method, and the one rule that governs it

A ruler is any rule that maps a cell's transcriptome to a porto-central coordinate. We compared many: Paper 2's exact 20+20 *landmark* genes; curated anchor sets; data-driven top-N and "full" sets ranked from Paper 2's zonation table; a *learned* dense PCA axis; sparse *lasso*/*elastic-net* axes; and a supervised whole-transcriptome classifier. Each yields coord = mean_z(PC) − mean_z(PP) or a learned projection.

The governing rule: ruler quality is judged ONLY on the healthy atlas — an 8-marker sign gate (pericentral CYP2E1/CYP1A2/GLUL/PCK2 must correlate +; periportal ASS1/ALDOB/PCK1/HAL −), the healthy PC–PP anti-correlation (a real axis has the two arms mutually exclusive), and split-half reproducibility. The best healthy ruler is *frozen* and only *then* applied to the disease cohort. Disease H1/H2/H3 are the *test*; they are never used to choose a ruler. Choosing by disease strength would be circular (leakage).

3 Results and analysis

3.1 Not every ruler is a ruler: a dilution curve

The 8-marker gate passes for almost everything (the markers are strong), but it is not sufficient. The decisive control is the *healthy PC–PP anti-correlation*. Small, high-SNR sets are genuinely bipolar (`paper2_landmark`: −0.45; `expanded_curated`: −0.48), but as the gene set grows the axis degrades and finally inverts: `paper2_top250` ≈ +0.04, `paper2_full` ≈ +0.55. Averaging ∼2000 weakly-zonated genes lets a shared technical/expression factor dominate, so the "full" coordinate is *not* a zonation axis at all. This is why a transcriptome-wide *unweighted* mean fails while a *weighted* or selected ruler succeeds.

ruler	PC	PP	healthy anticorr	split-half	H1 spread ρ	role
expanded_curated	26	23	-0.48	0.67	-0.43	primary_candidate
selected_frozen	26	23	-0.48	0.67	-0.43	primary_candidate
unsupervised_top100	100	100	-0.29	0.82	-0.61	primary_candidate
unsupervised_top50	50	50	-0.31	0.79	-0.62	primary_candidate
unsupervised	0	0	-0.31	0.78	-0.56	interpretability_only
paper2_landmark	20	20	-0.45	0.63	-0.38	primary_candidate
paper2_top50	50	50	-0.40	0.57	-0.32	primary_candidate
paper2_top100	100	100	-0.27	0.56	-0.28	primary_candidate
core_curated	13	8	-0.28	0.54	-0.54	interpretability_only
unsupervised_lasso	126	149	-0.18	0.63	-0.51	robustness
unsupervised_elasticnet	325	281	-0.07	0.69	-0.49	robustness
paper2_top250	250	250	0.04	0.55	-0.27	exploratory
paper2_full	1624	447	0.56	0.48	-0.04	exploratory
full	0	0	0.53	0.46	0.02	exploratory
supervised	0	0	-0.01	0.45	0.12	exploratory

3.2 Which ruler we selected, and why it is not the strongest-on-disease one

By healthy-ruler quality the selected primary set is `expanded_curated`. Notably this is *not* the set with the strongest disease trend (the small curated `core_curated` set has a steeper collapse) — choosing that would be cherry-picking. Selection used healthy metrics only; the disease result is the independent confirmation.

3.3 H1 — zonation collapses (and it is robust)

On every *valid* ruler, per-donor coordinate spread falls and the PC–PP anti-correlation rises toward zero across stages. The signal is concentrated and reproducible: landmark spread $\rho = -0.380$ (perm $p = 0.0085$); expanded $\rho = -0.434$; the label-free dense PCA axis $\rho = -0.562$ (perm $p = 0.0005$). The broken "full" ruler is null ($\rho = -0.042$), exactly as its healthy diagnostics predicted. The agreement of the published-signature ruler and a *label-free* ruler is the key point: the collapse is a property of the data, not of any one gene list.

3.4 Leakage control — the unsupervised axis is clean

A fair concern: was the learned axis trained on disease cells? No. It was learned on healthy cells only and frozen. To remove all doubt we also trained the axis *entirely on the external Paper 2 healthy atlas* (zero Paper 1 cells in fitting) and transferred it: the collapse survived ($\rho = -0.50$,

$p = 3 \times 10^{-4}$, 8/8 markers). And the label-free PCA axis correlates $\rho = +0.57$ with the landmark-signature coordinate ($\rho = +0.39$ with its PC arm, -0.45 with its PP arm) — so the axis is not a signature-formula artefact, and the result is not a variance-maximization artefact.

3.5 H2 — which programs lose zonation, and fastest

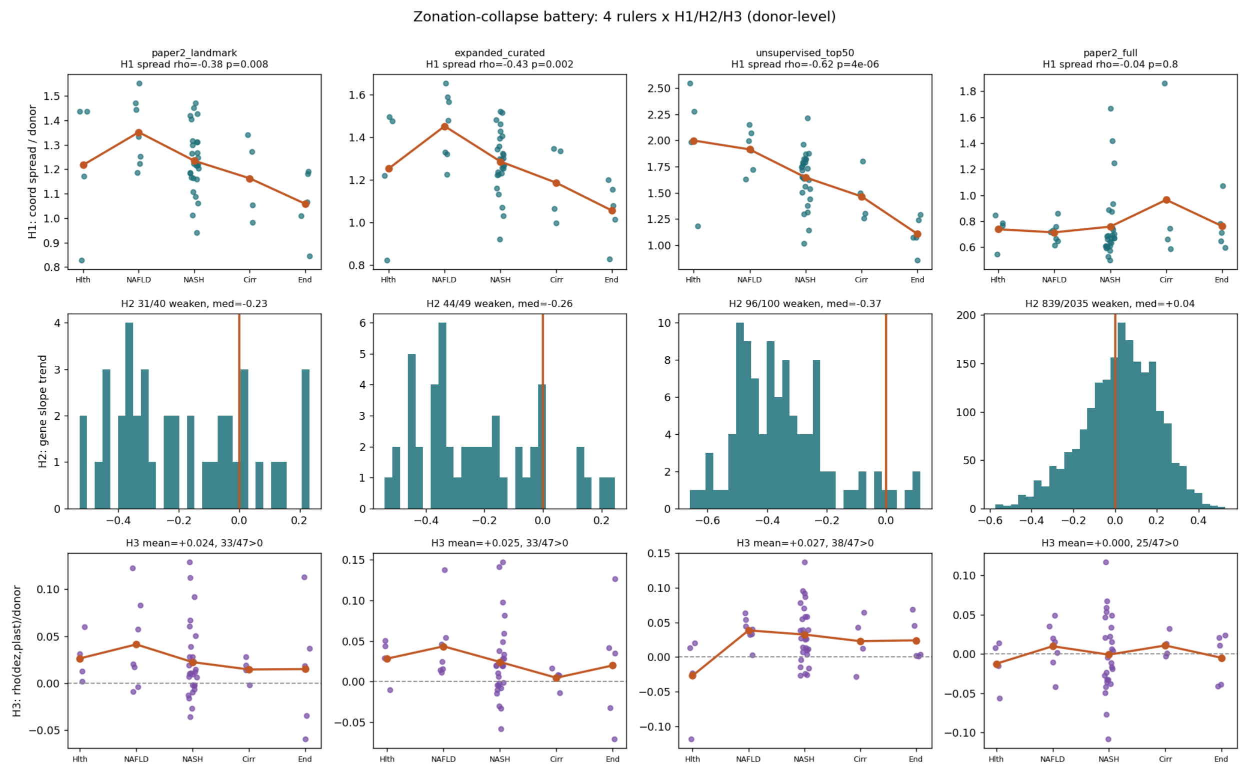
H2a (global): most zonated genes lose their donor-level zonal slope with disease (held-out split; e.g. 96/100 genes weaken for the unsupervised ruler). **H2b (differential vulnerability):** grouping genes into programs reveals a clear order — pericentral Wnt/identity genes de-zonate first, then nitrogen/urea and xenobiotic metabolism, whereas acute-phase/secreted and bile-acid programs are spared or even strengthen:

program	median slope-trend ρ	% weaken
wnt_pericentral_identity	-0.44	100%
urea_nitrogen	-0.29	58%
cyp_xenobiotic	-0.24	78%
lipid_metabolism	-0.03	50%
other	+0.04	41%
bile_acid_transport	+0.04	44%
cholangiocyte_plasticity	+0.06	50%
acute_phase_secreted	+0.09	32%

Biologically this is coherent: loss of the pericentral Wnt identity is the earliest and strongest de-zonation, while acute-phase genes change with disease *globally* (not zonally), so their zonal slope does not collapse. Note H2 (loss of spatial restriction) is distinct from classic DE (change in level); the cross-table `h2_slopeloss_vs_DE.csv` reports both per gene.

3.6 H3 — de-zonation couples weakly to plasticity

Within donors, de-zonation correlates positively with plasticity-marker expression (KRT7/KRT19/SOX9/SOX4/KI in ~70% of donors (landmark sign-test $p = 0.008$), but the effect is small (mean $\rho \approx 0.02$). This is the weakest of the three hypotheses: directionally consistent with Paper 1's transdifferentiation theme, but not a strong donor-level effect here.



4 Discussion

Three independent constructions — a published landmark signature, a label-free PCA axis (trained even on a fully external healthy atlas), and regularized sparse axes — converge on the same conclusion: hepatocyte zonation progressively collapses across MASLD, the pericentral Wnt-identity program leads the loss, and de-zonation is mildly associated with epithelial plasticity. Because ruler selection was confined to healthy-atlas criteria, the disease findings are genuine transfer results rather than fitted outcomes.

5 Limitations

(i) Donor numbers per stage are imbalanced (4/7/27/4/5), so cirrhosis/end-stage estimates are thin; all inference is donor-level with bootstrap CIs and permutation nulls to respect this. (ii) Paper 2 per-cell zone labels are an η-over-landmark reconstruction (Paper 2's own method) — spatially grounded but not a physical coordinate; classifier entropy is therefore auxiliary. (iii) The "full" unweighted set is a cautionary negative, not a result. (iv) H2b program groupings are curated and the smaller rulers contain few genes per program, so the gene-rich `paper2_full` table is used for the program comparison.

6 Conclusions

Hepatocyte zonation collapses with MASLD stage at the donor level (H1), driven first by the pericentral Wnt-identity program with nitrogen and xenobiotic metabolism following (H2), and is weakly coupled to epithelial plasticity (H3). The result is robust to how the ruler is built and to training the ruler on a fully external healthy atlas.

Companion artefacts: per-set dossier Zonation_Ruler_Report.pdf; machine tables signature_battery_summary.csv, selected_set_decision.txt, per-set results/tables/<set>/.